

Heeseung Choi

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Research Interests

My research interests include, but are not limited to:

- RF/Microwave Circuits & Front-End Systems
- Applied Electromagnetics
- Wireless Communication

Education

Yonsei University, Seoul, South Korea

03/2020 – 02/2026

Bachelor of Engineering in Electrical and Electronic Engineering

GPA: 4.07/4.30

University of Georgia, GA, United States

08/2024 – 12/2024

Exchange Program

GPA: 4.00/4.00

Research Experience

Undergraduate Researcher

Seoul, South Korea

Yonsei University — Advisor: *Prof. Byung-Wook Min*

01/2025 – Present

- Led an undergraduate capstone on a 1-bit Phase Shifting Switch Module for Reconfigurable Intelligent Surface, optimizing transistor sizing and matching networks to achieve low insertion loss and a stable 180° phase shift over a wide bandwidth.
- Awarded the IEEE AP-S Undergraduate Research Scholarship; presented at KIEES (2025).
- Currently extending the 1-bit design to 1.5-/2-bit switch modules for higher phase resolution and improved beam control; optimizing phase accuracy, insertion loss, and an 8-channel layout for dual-polarized RIS integration, with prototypes entering fabrication.

Publications & Presentations

- [1] **Heeseung Choi** and Byung-Wook Min, “1-bit Phase-Shifting Switch Module for Reconfigurable Intelligent Surfaces,” *KIEES Summer Conference*, Pyeongchang, South Korea, Aug. 2025 (Poster Presentation).

Awards & Scholarships

- **IEEE AP-S Undergraduate Summer Research Scholarship**
IEEE Antennas and Propagation Society (IEEE AP-S)
Selected among 102 applicants based on a proposal titled “A Wideband Reconfigurable Intelligent Surface using 1-bit GaAs HEMT Phase Shifter”; awarded \$3,000.
- **Korea-U.S. Advanced Technology Exchange Scholarship**
Korea Institute for Advancement of Technology (KIAT)
Awarded to 90 top engineering students selected for U.S. exchange programs, based on academic excellence and potential; \$9,000 per semester.
- **Yonsei University Semiconductor Track Program**
Yonsei University
- **High Honors (Dean’s List):** 2020, 2023, 2024, 2025
Yonsei University

Skills

- **Tools:** ADS, HFSS, MATLAB, PSPICE/LTSPICE, Vivado
- **RF Measurement:** VNA, SA, VSG, Anechoic Chamber
- **Relevant Courses:** Electronic Circuits I/II, Electricity & Magnetism I/II, Microwave Engineering, Antenna Engineering, Communication Theory, Communication Systems, Digital Signal Processing